
My Data Literacy Project

(Replace this with your Project Title)

Luca^{*1} Erik Müller^{*2} Mattis^{*3} Maximilian Wernz^{*4}

Abstract

7777Put your abstract here. Abstracts typically start with a sentence motivating why the subject is interesting. Then mention the data, methodology or methods you are working with, and describe results.

1. Introduction

deutsche Medaillenkrise bei den Olympischen Sommerspielen 2024

7777Motivate the problem, situation or topic you decided to work on. Describe why it matters (is it of societal, economic, scientific value?). Outline the rest of the paper (use references, e.g. to Section 6: What kind of data you are working with, how you analyse it, and what kind of conclusion you reached. The point of the introduction is to make the reader want to read the rest of the paper.

2. Data and Methods

The data were retrieved from the official archive by the Deutscher Leichtathletik-Verband (DLV), which documents the top performances across the major track and field disciplines (including running events, hurdles and steeplechase, jumping events, throwing events, and combined events) from 2001 onward. The archive covers performances across gender (male, female) and age groups (under 14 up to elite).

As the data for each year, age group, and gender are stored in separate PDF files, we used *pdfplumber* in combination with specific regular expressions to handle formatting differences

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Project report for the “Data Literacy” course at the University of Tübingen, Winter 2025/26 (Module ML4201). Style template based on the [ICML style files 2025](#). Copyright 2025 by the author(s).

over years. Outliers due to entry inconsistencies in the original files (e.g., irregular time formats) were corrected when the intended values could be inferred from external sources. Minor data gaps were manually filled and non-recurrent distances (e.g., 7.5 km events in selected youth categories) were excluded, yielding a final dataset of 226,683 entries.

We standardized discipline names and birth years to a four-digit format to ensure consistency across the dataset. Performance marks were converted into numeric values (seconds for track events and meters for field events) to allow statistical analyses. To facilitate comparisons across events, performances were transformed using World Athletics scoring tables, which express results on a common point scale. To calculate the points for each performance, we used XXX

7777In this section, describe *what you did*. Roughly speaking, explain what data you worked with, how or from where it was collected, its structure and size. Explain your analysis, and any specific choices you made in it. Depending on the nature of your project, you may focus more or less on certain aspects. If you collected data yourself, explain the collection process in detail. If you downloaded data from the net, show an exploratory analysis that builds intuition for the data, and shows that you know the data well. If you are doing a custom analysis, explain how it works and why it is the right choice. If you are using a standard tool, it may still help to briefly outline it. Cite relevant works. You can use the `\citep` and `\citet` commands for this purpose ([MacKay, 2003](#)).

3. Results

Covid + Are German Athletes getting worse?

In this section outline your results. At this point, you are just stating the outcome of your analysis. You can highlight important aspects (“we observe a significantly higher value of x over y ”), but leave interpretation and opinion to the next section. This section absolutely *must* include at least two figures.

4. Discussion & Conclusion

Use this section to briefly summarize the entire text. Highlight limitations and problems, but also make clear statements where they are possible and supported by the analysis.

Contribution Statement

Explain here, in one sentence per person, what each group member contributed. For example, you could write: Max Mustermann collected and prepared data. Gabi Musterfrau and John Doe performed the data analysis. Jane Doe produced visualizations. All authors will jointly write the text of the report. Note that you, as a group, are collectively responsible for the report. Your contributions should be roughly equal in amount and difficulty.

Notes

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5. Introduction

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6. Data and Methods

NOTE: This is also Data Cleaning For the analysis of performance evolution and performance density evolution we made some restrictions to the dataset to have comparable and sufficient data. We analysed for the U18, U20 and Adult age groups, representing the most relevant age groups with consistent data. U23 didn't have enough data, also for

younger age groups. Also always used the top 35 Ranks for analysis to have enough data for all disciplines and age groups. A Group is defined as a combination of Gender x Age group x Discipline to ensure consistency with the original leaderboards.

We compared two time periods: Data Start period (2001-2004) and End data period (2021-2024). We used four year periods to exclude one-year variances.

MAYBE: Since we found no common analysis procedure for this kind of sports data, we defined the performance measurements ourselves.

To measure the overall performance improvement of a group, we use the Median Difference ($\Delta\tilde{x}_\%$) to quantify general group performance shifts and represent the typical performance at the elite level and prevent distortion through extreme outlier performances. The median of each period (p) is calculated from the respective performance pools (S_p) of the four year periods. The relative change between the start ($\tilde{x}_s$) and the end period ($\tilde{x}_e$) is defined as:

$$\Delta\tilde{x}_\% = \left(\frac{\tilde{x}_e - \tilde{x}_s}{\tilde{x}_s} \right) \times 100 \quad (1)$$

A positive $\Delta\tilde{x}_\%$ indicates an overall improvement in performance, while a negative $\Delta\tilde{x}_\%$ shows a decrease in performance scores. Statistical significance for the comparison of the medians is measured using the Mann-Witney U test. Without assuming a normal distribution, this test allows to evaluate whether the distributions of the two periods differ significantly.

To measure Performance Density, we used the performance spread (G_p) from annual top 3 performances to the bottom 3 performances (e.g. Rank 33-35) (PSG) of our dataset for the two time periods. Averaging for 3 athletes prevents single-athlete outlier problems. Averaging for top 3 reflects the international standard of success (Podium), while bottom 3 provides a stable and comparable baseline. We used the Spread to measure the width of the elite level performances and we use all possible informations (e.g. Ranks) of our dataset and have a measurement of polarization without assuming a normal distribution. Mathematically, the gap (G_p) for a period (p) is defined as the difference between the collective mean of the top 3 and bottom 3 performances recorded across all years (n) within that period.

$$G_p = \frac{1}{n} \sum_{t=1}^n \bar{x}_{t,\text{top3}} - \frac{1}{n} \sum_{t=1}^n \bar{x}_{t,\text{low3}} \quad (2)$$

Furthermore, the evolution of the performance gap ($\Delta G_\%$) between the start period (G_s) and end period (G_e) is calculated as

$$\Delta G_\% = \left(\frac{G_e - G_s}{G_s} \right) \times 100 \quad (3)$$

A positive $\Delta G_\%$ suggests a decrease in PSG, while a negative $\Delta G_\%$ suggests an increase in PD.

To assess the statistical significance of the change in PSG, we use a bootstrap resampling method with $B = 10,000$. Bootstrap is non-parametric and can handle the low sample size ($N=12$) well suiting our data distribution. We draw with replacement from the performance pools, we generated a distribution of the difference in gaps (ΔG_{obs}). We then calculate the 95% confidence interval (CI) for the difference in gaps. If the CI includes zero, the null hypothesis of no significance change in PSG cannot be rejected.

(MacKay, 2003).

FOR RANK Analysis:

- Calculation of the mean per rank per group per period - calculation of the difference of the two periods per rank per group

7. Results

Figure X shows a change in performance spread and development from 2021-2024 compared to 2001-2004 in many groups. A significant change in the median or PSG was found in 90 groups (84.90%). While the median significantly increases in about half (39 Groups) and worsens in the other half (38 Groups), this balance is not seen in PSG. A significant increase in the performance spread was observed in 49 groups (42.06%). However, a significant decrease was only observed in 7 groups (6.60%), all of which were women's groups. In contrast, a significant increase in PSG can be found in the male age groups: In the U20 70.58% (12 of 17 groups), in the U18 64.71% (11 of 17) and in the adults 40.00%. A gender difference is also evident when considering the median. Across all age groups, 51.00% (27 out of 52 groups) of the women's groups showed improvement. This improvement was observed in only 10.03% (11 out of 56 groups) of the men's groups. When considering the different discipline groups, a difference can be found between track and field disciplines. In running and sprinting disciplines, a significant improvement across all groups was 3.5 (14 to 4 Groups) times more frequent and 1.625 (13 to 8 Groups) times more frequent, respectively. In comparison, the throwing and jumping disciplines both deteriorated in 2.6 (13 to 5, both) times as many groups.

PLOT: RANK Evolution

- PLOT shows the rising discrepancy: first places are getting better, lower places worse - In general, but especially the M U18 and U20, while women - better over all places

8. Discussion & Conclusion

The results clearly demonstrate a generally increased performance spread between elite performance levels. While the absolute top group (ranks 1–5) has seen steady improvements over the last two decades, the performance of the lower elite group (ranks 25–35) has stagnated or even deteriorated, confirming the suspected polarization of performance. The effect is more pronounced in younger age groups than in adults. Possible explanations could lie in the general trend in membership numbers within the German Athletics Association (DLV). DLV membership figures (Lei) show a decline of 11.86% from 2005 to 2024, and even 26.60% in the 27-60 age group. This latter age group, in particular, often includes coaches who are leaving, thus impacting performance development. There is a particular shortage of coaches among young athletes (CITE Artikel oder so), which could explain the more pronounced performance decline. Since technical disciplines (Field events) in particular require closer coaching support, the more significant declines could also be explained by the departure of the coach.

However, this overall development is not as pronounced among women; in fact, some groups are even experiencing a reduction in performance spreads. Historically, the professionalization of women's athletics began significantly later than that of men. Particularly in recent years, there have been better structural opportunities to pursue the sport at a high level, as X et al. describe in their article. This is promising, especially with regard to the future performance of German women's athletics, as general performance improvements seem possible.

At least in the running disciplines, the general increase in the median can also be attributed to the development of carbon shoes, as (Willwacher et al.) also demonstrate.

- Zusammenhang mit Corona Daten

- LIMITATION: TOP 35!

... The increase in the performance spread indicates a structural change in organized athletics. A future focus should be placed on strengthening the grassroots level. Following the principle of "the grassroots supporting the elite," this will allow for more top performances and could bring German athletics closer to the world's elite.

Contribution Statement

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References

- das leichtathletik-portal - downloads. URL <https://www.leichtathletik.de/service/downloads>.
- MacKay, D. J. *Information theory, inference and learning algorithms*. Cambridge university press, 2003.
- Willwacher, S., Mai, P., Helwig, J., Hipper, M., Utku, B., and Robbin, J. *Does advanced footwear technology improve track and road racing performance? An explorative analysis based on the 100 best yearly performances in the world between 2010 and 2022*. doi: 10.51224/SRXIV.297.